

Lecture 11

September 29

Gauss...

Learning objectives

- Explain how Bayes connects equally likely counts, before any observations, with a uniform prior on the unknown success probability.
- Use Bayes' fair-value argument to derive the relationship between joint and conditional probabilities, identifying where the probability of the first event enters the calculation.
- Show why the arithmetic mean minimizes the sum of squared errors when estimating a common value from repeated observations.
- Explain why maximizing the likelihood under independent normal errors gives least squares, and identify the circularity in Gauss' use of the arithmetic mean to justify the normal error curve.
- Distinguish the law of large numbers from the central limit theorem.
- Explain how Laplace's choice of weights minimizes an estimate's variance under independent, mean-zero errors with equal variance, and show how this leads to the least-squares estimate.

Bayer and insufficient reason

$$X \sim \text{Binom}(n, \theta)$$

If I don't know θ
then all X outcomes are equally likely

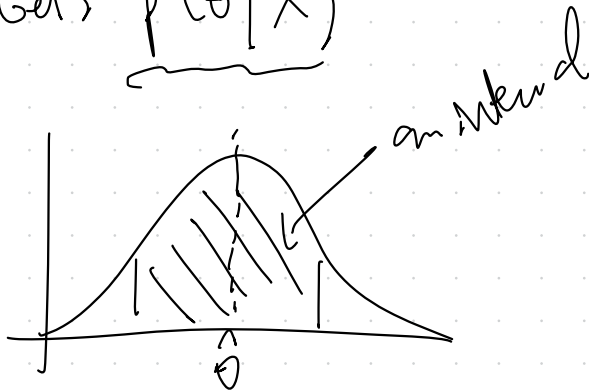
$$\Rightarrow p(\theta) = 1 \text{ for } \theta \in [0, 1]$$

prior is uniform

only true
if you assume

it true for all
 n , for all θ

$$\text{Get } p(\theta | X)$$



Bayes' conditional

Let $p = P(E_1)$, $q = P(E_2 | E_1)$, $r = P(E_1 \cap E_2)$

We get paid 1 if both events happen

Before the game you buy a ticket for r

If E_1 were to happen, ticket is worth $q - r \geq 0$

If E_1 were not to happen, ticket is worth 0, you lost r

Before the game, what is fair market value for the ticket after E_1 happens or doesn't happen

$$p(q - r) + (1 - p)(-r) = 0$$

$$pq - pr + pr - r = 0 \Rightarrow pq - r = 0$$

$$P(E_2 | E_1) = q = \frac{r}{p} = \frac{P(E_1 \cap E_2)}{P(E_1)}$$

Legendre's principle revisited

We observe $x_1, \dots, x_n$ and want to find m such that $\sum_i (x_i - m)^2$ is smallest

$$\sum (x_i - m)^2 = \sum (x_i^2 - 2mx_i + m^2)$$

$$\min_m \sum (x_i^2 - 2mx_i + m^2)$$

$$\frac{d}{dm} \sum (x_i^2 - 2mx_i + m^2) = \sum (-2x_i + 2m)$$

set deriv to zero

$$\sum (-2x_i + 2m) = 0$$

$$-2\sum x_i + 2\sum_{i=1}^n m = 0$$

$$-\sum x_i + nm = 0$$

$$\Rightarrow m = \frac{\sum x_i}{n}$$

Gauss goes for the errors

He wants an error curve that

- ① Has a max at zero
- ② Is symmetric
- ③ Is zero (ish) outside a range

(Laplace's curve
 $e^{-m|x|}$)

$$e^A e^B = e^{A+B}$$

The arithmetic mean is good,

Proof: The error curve that returns it is

$$\phi(\Delta) = \frac{h}{\sqrt{\pi}} \exp(-h^2 \Delta^2)$$

$$\Delta = \underbrace{x-m}_{\text{error}}$$

$$\pi = 3.14 \dots$$

h = precision

when maximized, splits out

$$m = \frac{\sum x_i}{n}$$

$$\begin{aligned} \phi(x_1 - m) \phi(x_2 - m) &= \frac{h}{\sqrt{\pi}} e^{-h^2(x_1 - m)^2} \frac{h}{\sqrt{\pi}} e^{-h^2(x_2 - m)^2} \\ &= \frac{h}{\sqrt{\pi}} \frac{h}{\sqrt{\pi}} \exp(-h^2[(x_1 - m)^2 + (x_2 - m)^2]) \end{aligned}$$

The normal distribution

Let $Y \sim N(\mu, \sigma^2)$: $E[Y] = \mu$, $\text{var}(Y) = \sigma^2$

$$\text{and } P(a < Y < b) = \int_a^b \frac{h}{\sqrt{2\pi}} \exp\left(-\frac{h^2}{2} (y-\mu)^2\right) dy$$
$$= \int_a^b \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} (y-\mu)^2\right) dy$$

$h = \frac{1}{\sigma}$, $h^2 = \frac{1}{\sigma^2}$ is the precision

Continuous

Takes on values between $-\infty$ and ∞

Laplace's CLT makes things less circular

$X_1, \dots, X_n$ are independent random variables

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty$$

$$P\left(\left|\frac{\sum X_i}{n} - \mu\right| > \varepsilon\right) \rightarrow 0$$

Prob. the $\frac{\sum X_i}{n}$ is
very different from $E[X_i]$
is small.

Law of Large Numbers

Uncertainty around $\left(\frac{\sum X_i}{n} - \mu\right)$?

$$\sqrt{n} \left(\frac{\sum X_i}{n} - \mu\right) \rightarrow N(0, \sigma^2)$$

Central Limit Theorem

How is the CLT relevant?

If errors are averages, then CLT says errors are Normal

lets try more Least Squares

We observe pairs $(x_1, y_1), \dots, (x_n, y_n)$ and want to relate them as

Find z such that $\sum (y_i - x_i z)^2$ is smallest

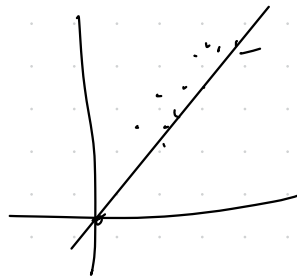
$$\min_z \sum (y_i^2 - 2x_i y_i z + x_i^2 z^2)$$

$$\frac{d}{dz} \sum (y_i^2 - 2x_i y_i z + x_i^2 z^2) = \sum -2x_i y_i + 2x_i^2 z$$

Set equal to zero

$$-2 \sum x_i y_i = -2z \sum x_i^2$$

$$z = \frac{\sum x_i y_i}{\sum x_i^2}$$



If errors are "just" mean 0 and symmetric?

Write the same problem as $\underline{z_i = y_i - x_i z}$
Laplace knows sums!

maybe: $\sum m_i z_i = 0$

$$\sum m_i z_i = \sum m_i y_i - \sum m_i x_i z \quad \text{where } m_i > 0 \text{ is a weight}$$

$$0 = \sum m_i y_i - z \sum m_i x_i$$
$$\Rightarrow z = \frac{\sum m_i y_i}{\sum m_i x_i}$$

Not 0:

$$z = \frac{\sum m_i y_i}{\sum m_i x_i}$$

+ u
↑
error term

$$u = \frac{\sum m_i z_i}{\sum m_i x_i}$$

average of errors

$$z = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{\sum x_i y_i}{\sum x_i^2}$$

uses CLT to show $u \approx N(\text{mean}, \text{var})$

$$\underline{\text{var}(u)} = \text{var}\left(\frac{\sum m_i \varepsilon_i}{\sum m_i x_i}\right)$$

$$= \left(\frac{1}{\sum m_i x_i}\right)^2 \text{var}\left(\sum m_i \varepsilon_i\right)$$

$$= \frac{1}{(\sum m_i x_i)^2} \sum \text{var}(m_i \varepsilon_i) \quad \text{by independence}$$

$$= \frac{\sum m_i^2 \text{var}(\varepsilon_i)}{(\sum m_i x_i)^2} \leftarrow \text{the same}$$

$\propto \frac{\sum m_i^2}{(\sum m_i x_i)^2}$
proportional
to

To find z I need to
choose m_i
When you minimize $\frac{\sum m_i^2}{(\sum m_i x_i)^2}$
in m_i you'll find
 $m_i \propto x_i$